



CORIOLIS

TECHNOLOGIES

Summer Internship Report

Author: Divyanshi Kumari

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Subjective Question & Answer Generator from PDF & Evaluator

1. Abstract

The objective of this project is to develop an application leveraging LangChain capabilities to automate the generation of subjective questions and answers from PDF documents and to evaluate student responses. The application simplifies the extraction of information and the generation of answers while focusing on producing a diverse range of question types, including situational, problem-solving, and factual questions. Additionally, it offers rigorous evaluation mechanisms to assess the quality of student responses. Through this project, we aim to address a critical need in education and assessment by automating time-consuming tasks and enhancing the quality of question generation and evaluation processes.

2. Problem Statement

The problem at hand is to automate the generation of subjective questions and answers from PDF documents and to develop an evaluation mechanism for assessing user responses. The system should efficiently generate diverse types of questions and ensure both accurate question and answer generation, as well as effective evaluation of responses.

3. Workflow

The workflow for developing a subjective question and answer generator & evaluator includes the following stages:

1. **Data Collection:** Gather source materials, such as PDFs or text documents.
2. **Data Preprocessing:** Extract and parse text from the collected documents, then chunk the text into manageable components.
3. **Question Generation:** Create a variety of question types based on the processed text chunks.
4. **Answer Generation:** Generate answers for the questions using the most relevant text chunks.
5. **Evaluation:** Assess the relevance and accuracy of the user answers.

3.1 Data

Here's a brief overview of the data sources:

1. **Private Website:** <https://discourse.lan.coriolis.co.in/> - Features internal articles

and posts on guidelines for employees, terms, and policies at Coriolis- Policies on Moonlighting, helmet use, Core Values, Sexual Harassment policies etc. Access requires Coriolis VPN and credentials.

2.Public Sources: Includes publicly available [Amazon Brand Protection report](#), machine learning research papers, university prescribed books.

3.2 Data Preprocessing & Chunking

The PDF is first read using PdfReader, from PyPDF2 library, which provides access to the content of each page. The text from all pages is then extracted and combined into a single string. To manage this text more effectively, RecursiveCharacterTextSplitter is employed to divide the text into chunks of a specified size, with a 100-character overlap between adjacent chunks. This approach ensures that the text is segmented in a manner that preserves contextual continuity and facilitates easier handling in subsequent processing steps. The outcome is a list of text chunks, which can be utilised for further analysis or specific task-related operations.

3.3 Question Generation

We started with a basic prompt that required the chunked pdf and user specified number of questions to generate questions, however the results were pretty basic what and how questions as you can see in the snippet.

```
' 1. How can machine learning address problems where human experts are unable to explain their expertise?',
' 2. What is the difference between analytical and empirical learning? Provide an example of each.',
' 3. How can a machine learning system learn to predict machine failures in modern automated manufacturing facilities?',
' 4. What are the advantages and disadvantages of decision trees as a machine learning algorithm?',
' 5. How can a machine learning system learn to predict the future behavior of the stock market?',
' 6. How can a machine learning system learn to filter unwanted electronic mail messages for each computer user separately?',
' 7. How can a machine learning system learn to play optimal chess?',
' 8. What is the difference between bias and variance in machine learning?',
' 9. How can a machine learning system learn to recognize speech?',
' 10. How can a machine learning system learn to diagnose breast cancer from medical images?']]
```

3.3.1 Modifications:

- **Prompt Adjustments:** During our ongoing evaluation under Giskard RAGET, we identified a diverse range of question types that can be generated from PDFs. This inspired us to refine our prompt strategy to better align with our core functionality of generating questions that can be answered directly from the same PDFs. The updated question types include Situational, Problem Solving, Inference based, Comparative Analysis, Descriptive, Factual.

By integrating these question types, we aim to enhance the breadth and depth of questions generated, thereby improving the overall effectiveness of our PDF-based question generation process.

- **Performance Checks:** Evaluated the retriever's performance and answer quality using Giskard RAGET.

3.3.2 Improvements:

QUESTION : 18. Problem-Solving: If you are facing issues with a high misclassification rate in a medical diagnosis classifier, what steps would you take to address this problem?

QUESTION : 17. Situational: Suppose you are developing a model for predicting the El Niño phenomenon. What kind of data would you need, and how would you structure your learning algorithm?

QUESTION : 14. Descriptive: Explain the process of how a decision tree is constructed according to the text.

QUESTION : 13. Fact-Based: What is a labeled example in the context of supervised learning?

QUESTION : 5. Inference Questions: Based on the text, why is it important to have a separate test set when evaluating the performance of a machine learning classifier?

QUESTION : 2. Difference-Based: Compare and contrast empirical learning and analytical learning as defined in the document. Provide examples of each.

3.4 Context Retrieval

The BM25 Retriever from LangChain is used to extract relevant documents from chunked PDFs by leveraging the BM25 model. This established information retrieval technique excels at processing and ranking documents based on the occurrence and rarity of query terms. When questions are derived from the content of the PDF itself, BM25's term-based ranking aligns well with the terms in the questions and the document. This alignment enhances the likelihood that the retrieved information will directly address the query, thereby improving the relevance and accuracy of the results.

3.5 Answer Generation

To generate answers, I have created a custom processing chain using LangChain. I have set up a chain that includes applying the prompt, passing the context and question into it, invoking the language model to generate an answer. Inspired by Retrieval-Augmented Generation (RAG) principles, this approach ensures that the responses are accurate and contextually aligned with the retrieved information from the PDF.

3.6 Evaluation

We aimed to develop an evaluation mechanism that meets human standards by effectively identifying issues such as incompleteness, unsupported information, vagueness, and improper responses. We began by utilising **Langsmith**, which offered metrics like "answer-hallucination," "answer-helpfulness," and "correctness." Inspired by these prompts, we created our own local evaluation system, which does

not rely on Langsmith but incorporates our own set of parameters. We settled on evaluating answers based on accuracy(check for no hallucination), completeness, clarity, and helpfulness.This approach allows us to detect incomplete or partially complete answers, responses that are missing from the document, and those that might simply be "I don't know," while ensuring that high-quality answers are properly recognized and assessed.

3.6.1 APP Generated Questions:

Below(Q.5) is a snippet of a good answer evaluation.

Question: 5. Inference Questions: Based on the text, why is it important to have a separate test set when evaluating the performance of a machine learning classifier?

Student Answer: It is important to have a separate test set when evaluating the performance of a machine learning classifier to ensure that the evaluation is unbiased and reflects the model's ability to generalize to new, unseen data. A separate test set helps to assess how well the model performs on data it has not been trained on, providing a true measure of its predictive accuracy and preventing overfitting to the training data.

	accuracy	completeness	clarity	helpfulness
score	10	8	9	9
justification	The answer accurately reflects the information provided in the document regarding the importance of a separate test set in evaluating a machine learning classifier.	The answer covers the key points about the need for a separate test set to prevent overfitting and to measure generalization. However, it could have explicitly mentioned the potential for a classifier to perform well on training data but poorly on unseen data, which is a critical aspect discussed in the document.	The answer is clear and easy to understand, using straightforward language. However, a slight improvement could be made by breaking down the concepts into simpler terms for broader accessibility.	The answer directly addresses the question and provides relevant information. It could be slightly more helpful by including a brief example or context from the document to illustrate the points made.

FINAL SCORE

Score: 9

Justification: The answer is strong in accuracy and clarity, with minor gaps in completeness and helpfulness. It effectively addresses the question while maintaining a high level of relevance to the document.

Below (Q.2), is a snippet of an incomplete but correct answer evaluation. The question demands a difference between empirical and analytical learning but the user has only mentioned empirical learning.

Question: 2. Difference-Based: Compare and contrast empirical learning and analytical learning as defined in the document. Provide examples of each.

Student Answer: Empirical learning and analytical learning differ in their approaches to acquiring knowledge.

Empirical Learning is based on direct experience and observation. It involves learning through practical engagement and experimentation.

	accuracy	completeness	clarity	helpfulness
score	5	3	6	4
justification	The answer correctly identifies that empirical learning is based on direct experience and observation, but it does not mention analytical learning or provide a clear definition or example of it, which is a significant omission.	The answer fails to cover key points from the document, such as the definition of analytical learning, the distinction between the two types of learning, and specific examples provided in the document. It only partially addresses empirical learning.	The answer is relatively clear in its explanation of empirical learning, but it lacks clarity regarding analytical learning due to its absence in the response.	The answer does not fully address the question of comparing and contrasting empirical and analytical learning, as it only discusses empirical learning without providing a comparison or examples of analytical learning.

FINAL SCORE

Score: 4

Justification: The answer lacks completeness and helpfulness, as it does not adequately address the question of comparing and contrasting the two types of learning. While it contains some accurate information about empirical learning, the absence of analytical learning significantly detracts from its overall effectiveness.

Below (**Q.18**), you can see another bad answer, the user has written I don't know for a question which can be answered from the PDF.

Question: 18. Problem-Solving: If you are facing issues with a high misclassification rate in a medical diagnosis classifier, what steps would you take to address this problem?

Student Answer: I don't know

	accuracy	completeness	clarity	helpfulness
score	1	1	5	1
justification	The answer 'I don't know' does not provide any information grounded in the document and is considered hallucination.	The answer fails to address any key points from the document regarding steps to take for reducing misclassification rates in a medical diagnosis classifier.	While the answer is clear in its expression of uncertainty, it does not provide any useful information or context.	The answer does not address the question at all, providing no relevant information or guidance on how to solve the problem of high misclassification rates.

FINAL SCORE

Score: 1

Justification: The answer is unhelpful, inaccurate, and incomplete, failing to provide any relevant information or steps to address the problem posed in the question.

3.6.2 User Based Questions:

- Apart from evaluating user answers to APP-Generated Questions, the project also facilitates evaluation on **USER Based Questions**. The answers are only

evaluated if the queries are relevant to the pdf. If found relevant, answers are evaluated on the basis of uploaded pdf.

The topic supervised learning is relevant to a pdf on foundations of machine learning.

Question: What is supervised learning?

Student Answer: Supervised learning is a type of machine learning where the model is trained on a dataset that includes both input features and their corresponding labeled outputs. The goal is for the model to learn a mapping from inputs to outputs so that it can make accurate predictions on new, unseen data. The training process involves minimizing a loss function that measures the difference between the model's predictions and the actual outputs. Examples include classification (e.g., spam detection) and regression (e.g., predicting house prices)

	accuracy	completeness	clarity	helpfulness
score	9	6	8	7
justification	The answer accurately describes supervised learning as a type of machine learning that involves training on labeled datasets. However, the mention of specific examples like spam detection and predicting house prices, while relevant, is not explicitly supported by the document.	The answer covers the basic definition of supervised learning and its goal, but it misses several key points from the document, such as the importance of generalization, the role of training and test sets, and the distinction between classification and regression tasks. These are critical aspects of supervised learning that were highlighted in the document.	The answer is generally clear and easy to understand, using straightforward language. However, the explanation of minimizing a loss function could be simplified for better clarity.	The answer addresses the question of what supervised learning is, but it could be more directly tied to the specific content of the document. The examples provided, while helpful, may distract from the core definition and key points outlined in the document.

FINAL SCORE

Score: 7

Justification: The answer provides a solid overview of supervised learning but lacks completeness in covering all key points from the document. While it is accurate and clear, the missing details significantly impact its overall effectiveness in fully addressing the question.

Since the uploaded pdf is a foundation on machine learning with no mention of Generative AI. The below provided snippet shows an error that the question might be irrelevant.

Evaluation on User Provided Questions

Enter number of questions

1

- +

Your Question 1:

What is Generative AI?

Your Answer to Your Question 1:

Generative artificial intelligence is artificial intelligence capable of generating text, images, videos, or other data using generative models, often in response to prompts. Generative AI models learn the patterns and structure of their input training data and then generate new data that has similar characteristics.

Evaluate User Answers

The questions seems not completely relevant to the text in pdf. Try Going through the PDF again. Try framing the question properly and re run.

3.6.3 Reference based evaluation:

- In case, the user does not have a PDF , the user can proceed to **Reference answer based evaluation** which allows the user to input a benchmark response to evaluate your answer to.

QUESTION : What is a confusion matrix? State some metrics to evaluate your ML model.

REFERENCE ANSWER: A confusion matrix is a performance evaluation tool for classification models. It's a table that summarizes the performance of a classification algorithm on a set of test data. It displays the number of correct and incorrect predictions made by the model. Components of a Confusion Matrix: True Positive (TP): Correctly predicted positive cases. True Negative (TN): Correctly predicted negative cases. False Positive (FP): Incorrectly predicted positive cases (Type I error). False Negative (FN): Incorrectly predicted negative cases (Type II error). Several metrics can be derived from a confusion matrix to evaluate a model's performance: Accuracy Ratio of correct predictions to total predictions. $\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$ Precision Proportion of positive predictions that were actually correct. $\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$ Recall (Sensitivity) Proportion of actual positive cases that were correctly identified. $\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$ F1-Score Harmonic mean of precision and recall. $\text{F1-Score} = 2 * \frac{(\text{Precision} * \text{Recall})}{(\text{Precision} + \text{Recall})}$ Other Metrics Specificity: Proportion of actual negative cases that were correctly identified. ROC Curve: Plots the true positive rate against the false positive rate. AUC (Area Under the ROC Curve): Measures the overall performance of a classification model

STUDENT ANSWER: Confusion Matrix: A confusion matrix is a table used to evaluate the performance of a classification model. It summarizes the results of predictions by comparing the actual labels with predicted labels. It typically includes: True Positives (TP): Correctly predicted positive cases. True Negatives (TN): Correctly predicted negative cases. False Positives (FP): Incorrectly predicted positive cases (Type I error). False Negatives (FN): Incorrectly predicted negative cases (Type II error).

The model can be evaluated on : Accuracy, precision,recall,F1 Score, ROC-AUC curve, Specificity

	accuracy	completeness	clarity	helpfulness
score	10	8	9	9
justification	The answer accurately reflects the information provided in the document without introducing any unsupported claims or hallucinations.	The answer covers the key components of a confusion matrix and lists several evaluation metrics. However, it does not explain the formulas for the metrics, which are important for a comprehensive understanding.	The answer is mostly clear and easy to understand, but the list of metrics could be formatted better for readability.	The answer directly addresses the question about confusion matrices and evaluation metrics, but it could provide more context or explanation for the listed metrics.

FINAL SCORE

Score: 8.5

Justification: The answer is accurate and mostly complete, but it lacks detailed explanations of the metrics, which affects its overall helpfulness. The clarity is good, but slight improvements in formatting could enhance readability.

3.6.4 Comparison of Prompt-Based Scores with Human Evaluations

To validate the accuracy and effectiveness of our automated evaluation mechanism, a comparative analysis of the scores assigned to individual answers by our prompt-based evaluation with those provided by human evaluators is done. This analysis aims to confirm that our evaluation method aligns with human judgement and effectively measures answer quality. By aligning automated scores with human assessments, we evaluated the consistency and reliability of our scoring system.

4. Large Language Models

Large Language Models (LLMs) are advanced deep learning models pre-trained on extensive datasets using transformer architecture, which includes encoders and decoders with self-attention mechanisms. This setup allows LLMs to understand and generate text by capturing complex relationships between words.

Unlike domain-specific models, which are costly and less efficient, LLMs offer a versatile solution for various tasks in natural language processing (NLP) and artificial intelligence (AI). Notable examples include OpenAI's ChatGPT-3 and GPT-4, Meta's Llama, Google's BERT/RoBERTa and PaLM, and IBM's Granite series on watsonx.ai, which powers IBM's products.

LLM worked with: `Mistral-7B-Instruct-v0.2, gpt-3.5-turbo, gpt-4o-mini`

To enhance the project's efficiency and performance, a strategic transition between language models was implemented. Initially utilising `Mistral-7B` from HuggingFace, challenges arose in terms of processing speed and resource consumption. Subsequently, OpenAI `gpt-3.5-turbo` was adopted to address these limitations. However, constraints related to context length impacted the model's effectiveness when handling large PDF documents. The integration of `gpt-4o-mini` as the **primary language model** optimised the system by providing faster processing, increased context length, and reduced costs, ultimately improving the overall performance and efficiency of the project.

5. Results of Question-Answers generated

QUESTION : Context-Based: What are the four general categories of tasks for which machine learning is particularly useful, as described in the text?

ANSWER : The four general categories of tasks for which machine learning is particularly useful, as described in the text, are:

Problems for which there exist no human experts.

Problems where human experts exist, but they are unable to explain their expertise.

Problems where phenomena are changing rapidly.

Applications that need to be customised for each computer user separately.

QUESTION : Situational: Imagine you are developing a machine learning system to predict machine failures in a manufacturing facility. What data would you collect, and how would you approach the learning task?

ANSWER : To develop a machine learning system to predict machine failures in a manufacturing facility, you would collect the following types of data:

Sensor Readings: Continuous data from various sensors on the machines, such as temperature, vibration, pressure, and operational speed. This data can provide insights into the machine's current state and performance.

Historical Failure Data: Records of past machine failures, including timestamps, failure modes, and any relevant operational conditions leading up to the failure. This data is crucial for understanding the patterns that precede failures.

Maintenance Records: Information on maintenance activities performed on the machines, including dates, types of maintenance, and any parts replaced. This can help identify the impact of maintenance on machine performance.

Operational Data: Data on the operational context, such as production schedules, load conditions, and environmental factors (e.g., humidity, temperature in the facility).

Anomaly Data: Instances of unusual behaviour or anomalies detected in the machines, even if they did not lead to failures. This can help the model learn to recognize early warning signs.

Approach to the learning task:

Data Preprocessing: Clean and preprocess the collected data to handle missing values, normalise sensor readings, and convert categorical data into numerical formats if necessary.

Feature Engineering: Extract relevant features from the raw data that may indicate impending failures. This could include statistical measures (e.g., mean, variance) of sensor readings over time, trends, and changes in operational patterns.

Labelling: Create labelled examples for supervised learning by marking instances of sensor data leading up to a failure as positive examples and normal operational data as negative examples.

Model Selection: Choose appropriate machine learning algorithms for the task, such as decision trees, random forests, or neural networks, that can handle time-series data and learn from the features extracted.

Training and Validation: Split the data into training and validation sets. Train the model on the training set and evaluate its performance on the validation set using metrics such as accuracy, precision, recall, and F1 score.

Deployment and Monitoring: Once the model is trained and validated, deploy it in the manufacturing environment. Continuously monitor its performance and update the model with new data to improve its predictive capabilities over time.

By following this approach, the machine learning system can learn to predict machine failures effectively, allowing for proactive maintenance and minimising downtime in the manufacturing facility.

QUESTION : Problem-Solving: If a machine learning algorithm is overfitting the training data, what strategies could you employ to improve its generalisation accuracy?

ANSWER : To improve the generalisation accuracy of a machine learning algorithm that is overfitting the training data, you could employ the following strategies:

Pruning: In decision trees, for example, pruning procedures can be used to remove branches from an overly large tree, which helps reduce the risk of overfitting.

Regularisation: For algorithms like neural networks, support vector machines, and linear discriminant functions, regularisation methods can be applied. These methods impose a numerical penalty on having large numerical weights, effectively limiting the complexity of the resulting classifiers.

Increase Training Data: Providing more training data can help the algorithm learn better and generalise more effectively, as it will have more examples to learn from.

Simplify the Model: Choosing a simpler model with fewer parameters can help reduce overfitting, as simpler models are less likely to capture noise in the training data.

Cross-Validation: Using techniques like k-fold cross-validation can help in assessing the model's performance on unseen data and can guide adjustments to improve generalisation.

Early Stopping: In iterative algorithms, monitoring performance on a validation set and stopping training when performance begins to degrade can prevent overfitting.

Feature Selection: Reducing the number of features by selecting only the most relevant ones can help in reducing the complexity of the model and improve generalisation.

These strategies aim to balance the trade-off between bias and variance, ultimately leading to better generalisation on new, unseen data.

QUESTION: Fact-Based: What is the role of labelled examples in supervised learning, and how do they contribute to the learning process?

ANSWER : Labelled examples play a crucial role in supervised learning as they provide the necessary input-output pairs that the learning algorithm uses to construct a classifier. Each labelled example consists of an object (such as a photograph) and its associated class (such as male or female). The set of labelled examples provided to the learning algorithm is called the training set. The learning process involves analysing these examples to identify patterns and relationships that allow the algorithm to generalise and make accurate predictions on new, unseen data. The quality of the learned classifier is typically evaluated using a separate set of labelled examples called the test set, which helps measure the classifier's ability to generalise beyond the training data.

QUESTION : Difference-Based: Contrast supervised learning with unsupervised learning, providing examples of tasks that fall under each category.

ANSWER : Supervised learning and unsupervised learning are two distinct categories of machine learning tasks.

Supervised Learning: In supervised learning, the algorithm is trained on a labelled dataset, which means that each training example is paired with an output label. The goal is to learn a mapping from inputs to outputs so that the model can predict the output for new, unseen data. Examples of tasks that fall under supervised learning include:

Classification: For instance, developing a program that can determine whether a given photograph is of a male or female person based on labelled training examples.

Regression: An example would be predicting a person's age, height, or weight based on their photograph, where the output is a numerical value.

Unsupervised Learning: In contrast, unsupervised learning involves training on a dataset without labelled outputs. The algorithm tries to learn the underlying structure or distribution of the data without any specific guidance on what to predict. Examples of tasks that fall under unsupervised learning include:

Clustering: Grouping a set of objects into clusters based on their similarities without predefined labels.

Dimensionality Reduction: Techniques like Principal Component Analysis (PCA) that reduce the number of features in a dataset while preserving its essential characteristics.

In summary, supervised learning relies on labelled data to make predictions, while unsupervised learning seeks to find patterns or structures in unlabeled data.

Documentation-type pdf: Moonlighting

QUESTION : Inference Questions: Based on the text, what can be inferred about the company's stance on open source contributions, and under what conditions might they be acceptable?

ANSWER : The company's stance on open source contributions is generally positive, as they do not object to them in principle. However, contributions are acceptable only if they do not violate the company's or clients' intellectual property (IP) rights. Before making any open source contributions, employees are advised to check for potential business conflicts and seek written approval from management to ensure that their activities are authorised.

QUESTION : Fact-Based: What are the six consequences listed in the document that Alice would face if she is suspected of moonlighting?

ANSWER : 1. Alice will minimally lose her job for “unprofessional conduct.”

2. The relieving letter will state that Alice was terminated.
 3. Alice will be marked as a “do not re-hire” for posterity.
 4. Any subsequent HR reference check will reflect the above.
 5. If Alice is suspected of violating clients’ IP rights, the company is contractually obligated to inform the affected client(s) and cooperate with them in prosecuting her to the fullest extent of the law, if they choose to do so.
 6. The company would be liable for consequential damages to their clients and would have to prosecute Alice to recover costs and/or impose punitive damages.
-

QUESTION : Descriptive: Describe the types of activities that are generally considered acceptable for employees to engage in outside of their work at CTPL.

ANSWER : The types of activities that are generally considered acceptable for employees to engage in outside of their work at CTPL include:

Teaching music to neighborhood kids.

Volunteering at local organizations, such as a madrasa.

Building a website for their alma mater.

However, it is advised that employees check with management if they are in doubt about any outside activities. Additionally, open source contributions are generally acceptable as long as they do not violate company or client intellectual property rights.

QUESTION: Situational: If Alice is approached by a startup to work on a project that aligns with her interests but conflicts with CTPL's business, what steps should she take according to the guidelines provided?

ANSWER : According to the guidelines provided, if Alice is approached by a startup to work on a project that aligns with her interests but conflicts with CTPL's business, she should take the following steps:

Seek Written Approval: Alice must assume that any technical/software activity outside of Coriolis Technologies (CTPL) is unauthorized unless she seeks and obtains written approval from management beforehand.

Check for Business Conflicts: Before proceeding, Alice should check if there are any business conflicts with her current work at CTPL. This includes ensuring that her work for the startup does not violate CTPL's or its clients' intellectual property rights.

Consult Management: If Alice is in doubt about whether the project conflicts with CTPL's interests, she should send queries about potential business conflicts to the management team for clarification.

Respect Company Policies: If the project is deemed to conflict with CTPL's business interests, Alice should respect the company's policies and may need to consider her options, which could include leaving CTPL if she feels strongly about pursuing the project.

By following these steps, Alice can ensure that she is acting in accordance with the company's guidelines and protecting both her interests and those of CTPL.

6. Conclusion

This project highlights the transformative potential of large language models in automating content creation and evaluation. These models have been crucial in generating diverse and high-quality questions, while our evaluation mechanism has proven effective in assessing the accuracy and completeness of responses.

Comprehensive testing of evaluation prompts has confirmed the system's reliability and alignment with human judgement. By refining question generation and evaluation techniques and integrating advanced retrieval methods, the system is well-positioned to enhance its impact across academic, corporate, and research settings, ultimately improving user interactions and content analysis.

7. Appendix

1. Tools Used:

- 1.1. Langchain
- 1.2. OpenAI
- 1.3. Python
- 1.4. PyPDF2
- 1.5. Langsmith
- 1.6. Streamlit

2. References

- 2.1. Giskard RAGET  Giskard RAGET
- 2.2. Comparison on Human vs Prompt based Evaluation
 Evaluation on Test QA pairs